Measurement of matter-antimatter asymmetries with data from the LHCb experiment

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Abstract

The aim of this experiment is to understand and measure the differences in the behaviour of matter and antimatter with Run1 data recorded by the LHCb experiment. The three-body decay $B^{\pm} \to K^{\pm}K^{+}K^{-}$ is studied. An analysis on simulated data and experimental data to extract the invariant mass distribution was performed, followed by the determination of global and local matter-antimatter asymmetry, in search of CP violation effects in Standard Model processes.

Introduction

1 CP Violation

The study of matter-antimatter asymmetries, or Charge-Parity (CP) violation, is a fundamental aspect of modern particle physics. The CP transformation changes a particle in the respective anti-particle, so a violation of CP symmetry indicates a difference in the behavior of matter and antimatter. This can provide the key insights into the matter-antimatter asymmetrical production during the Big Bang. The measurements of CP violation obtained until now are predicted from the Standard Model but they do not fully explain the differences in matter and anti-matter we observe in our universe.

In the Standard Model (SM), the CP violation mechanism can be described by the Cabibbo-Kobayashi-Maskawa (CKM) matrix [1] as expressed below:

$$V_{\mathrm{CKM}} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix} = \begin{pmatrix} 1-\frac{1}{2}\lambda^2 & \lambda & A\lambda^3(\rho - i\eta) \\ -\lambda & 1-\frac{1}{2}\lambda^2 & A\lambda^2 \\ A\lambda^3(1-\rho-i\eta) & -A\lambda^2 & 1 \end{pmatrix} + \mathcal{O}(\lambda^4)$$

where λ, A, ρ and η are Wolfenstein parameters. The parameter η is responsible for CP violation. If $\eta = 0$, the matrix is purely real, and CP violation disappears entirely in the Standard Model.

B meson decays provide an excellent framework for studying CP violation. These processes involve the weak transition of a b-quark into lighter quarks ($u, c, s,$ or d), which can occur via both tree-level and loop-level (penguin) Feynman diagrams. In this experiment, we specifically focus on charmless decays because they are mediated by electroweak interactions where CP violation is prominent. For instance, at the tree level, a $b \to u\bar{u}s$ transition is proportional to the product of the CKM matrix elements $V_{ub}V_{us}^*$. Our focus here is on the V_{ub} element, since it is proportional to $(\rho - i\eta)$ in the Wolfenstein parameterization of the CKM matrix, so we can get access to the parameter related to CP violation. The Feynman diagram for the analyzed charmless three-body decay of a charged B meson into three charged kaons is shown in Figure 1.

2 LHCb detector

The LHCb experiment [2] is a sophisticated particle physics detector located at the Large Hadron Collider (LHC) at CERN. It is specifically designed to perform precision measurements to study the differences between matter and antimatter by investigating CP violation in rare decays of particles containing b (bottom) and c (charm) quarks.

LHCb is a single-arm forward spectrometer optimized to capture particles produced in the forward direction. This is crucial because b-hadrons are typically produced in the forward region with high longitudinal boost due to their

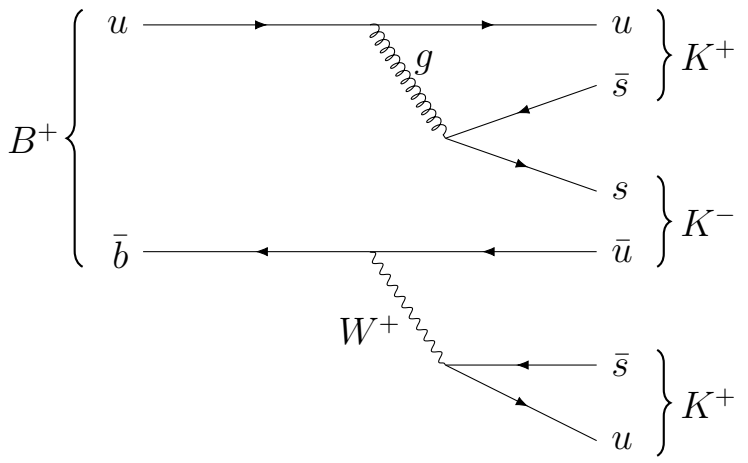


Figure 1: Feynman diagram of a representative three-body $B^+ \to K^+K^-K^+$ decay

momentum. A schematic view of the detector is shown in Figure 2. A right-handed coordinate system is used for description in the image: the z-axis is along the beam and the y-axis is vertical. The negative (positive) z-region of the detector is referred to as upstream (downstream).

The LHCb detector consists of several subsystems with different utilization;

- **Vertex Locator (VELO):** It is placed surrounding the interaction region, just millimeters away from the LHC beams. Composed of silicon pixel sensors, the VELO provides exceptional spatial resolution. It is critical for separating primary collision vertices from secondary vertices of short-lived beauty and charm states.

- **Dipole Magnet:** A large, non-superconducting conventional magnet generating an integrated magnetic field of approximately 4 Tm perpendicular to the horizontal bending plane. It deflects charged particles to enable momentum measurement and particle identification.

- **Tracking system:** The systems include the Trigger Tracker (TT or Tracker TUricensis) which is located upstream of the dipole magnet, and three tracking stations (T1, T2, T3) which are located downstream of the magnet. These components measure the momentum of charged particles by tracking their paths through the magnetic field, which bends the trajectories of charged particles depending on their momentum. The tracking stations also consist of drift tubes filled with gas which generate signal from ionization.

- **RICH detectors:** The RICH detectors distinguish charged hadrons (pions, kaons, and protons) by utilizing the Cherenkov effect, where particles

traveling faster than the speed of light in a medium emit Cherenkov radiation. The RICH detectors, using different radiators (aerogel, C4F10, CF4), cover a wide range of momenta and provide high particle identification efficiency. By measuring the angle of Cherenkov radiation and combining it with momentum information, the particle's mass and type can be determined.

- **Calorimeter:** The full calorimeter system consists of an electromagnetic calorimeter (ECAL), a hadronic calorimeter (HCAL), a scintillating pad detector (SPD), and a preshower detector (PS). These devices measure the energy of particles by absorbing them and producing scintillation light, which is proportional to the energy deposited. ECAL is optimized for detecting photons and electrons, while HCAL is designed for hadrons. These measurements are essential for identifying particles and understanding their energy distributions.

- **Muon system:** The muon system comprises multiple layers of muon chambers alternated with iron absorbers. This configuration ensures high-efficiency muon identification, which is essential for analyzing decay processes involving muons, which are the minimum ionization particle (MIP) at LHC.

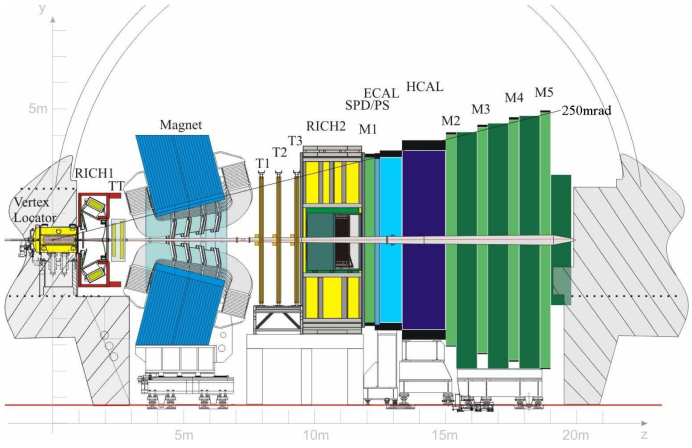


Figure 2: Schematic overview of the LHCb detector and its subsystems.

3 Analysis

This section details the analytical steps taken to investigate charge-parity (CP) violation in charmless $B^{\pm} \to K^{\pm}K^{\mp}K^{\pm}$ decays using both simulated and real data from Run 1 of the LHCb experiment.

First, we reconstruct the invariant mass of the B meson candidates; simulated data is initially used to validate the kinematics, followed by the application of strict particle identification (PID) pre-selection cuts on real data to successfully isolate the signal from the combinatorial background. Next, we perform a global CP asymmetry search by comparing the total decay rates of B^+ and B^- mesons to look for overall matter-antimatter differences. Finally, to achieve a more sensitive and targeted measurement, we study the internal decay dynamics using Dalitz plots. This involves applying a mass-ordering technique (folded Dalitz plots), identifying and vetoing intermediate background resonances (such as the charm meson D^0), and comparing the B^+ and B^- kinematic distributions. By mapping the local asymmetry and its statistical significance, we isolate the specific kinematic region where CP violation is most prominent.

For the data analysis, we used data from the first run recorded in 2011 by the LHCb experiment, during the first major data-taking period of the LHC. The data to be analysed in this project comes from collisions of two proton beams accelerated at LHC to a centre-of-mass energy of 7 TeV. Each of these beams consists of about 2000 bunches, each containing about 1.1×10^{11} protons. These bunches have been brought into collision within the LHCb detector at an average rate of 15 MHz.

3.1 Dataset

The dataset contains 26 features that describe the flight distance of the B mesons from the primary vertex and its quality of vertex fit. It also consists of reconstructed momentum, charges, and the identification probability for kaons, pions and muons of the three hadrons from the decay product. These features were used to perform the preselection of the $B^{\pm} \to K^{\pm}K^{+}K^{-}$ decay channel, to reduce the background specifically. The preselection cuts are explained in the next section.

3.2 Invariant Mass Reconstruction

The invariant mass is a characteristic property of each particle. Particle identification can be performed efficiently by using their mass information. The mass of a particle can be calculated using the energy-momentum relation from special relativity:

$$E^2 = p^2 + m^2 \tag{1}$$

Since the dataset includes the momenta of the daughter particles (kaons), the invariant mass of the B meson can be reconstructed through the conservation of energy and momentum.

The analysis began with an exploration of the simulated data. Using the known rest mass of the kaon and the momentum components provided by the simulation, we first computed the individual energy of each daughter kaon. By summing the energies and momenta of the three kaons, we reconstructed the total four-momentum of the parent B meson. Finally, we calculated the

invariant mass of the B meson candidates and visualized its distribution in a histogram, expecting a sharp peak around the true B meson mass.

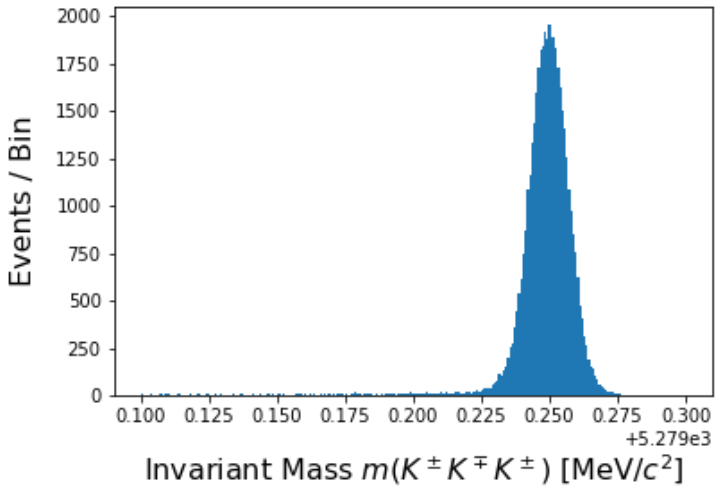
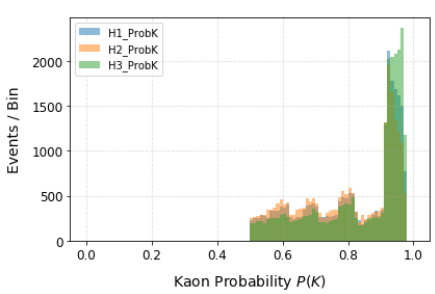


Figure 3: Reconstructed invariant mass distribution for the three kaons system, with the simulated data.

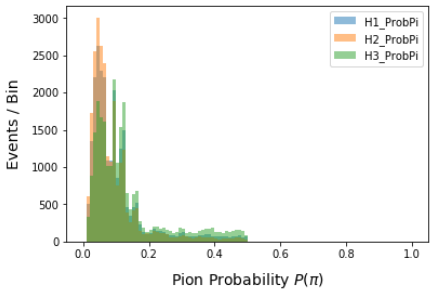
As shown in Figure 3, the reconstructed invariant mass distribution for the simulated data exhibits a sharp peak at 5279.25 MeV/c^2. This value perfectly matches the nominal mass of the charged B meson as reported by the Particle Data Group (PDG) [3].

Following the validation on simulated data, the same kinematic reconstruction procedure is applied to the real dataset. However, unlike the pure simulation, raw detector data contains a significant amount of combinatorial background and misidentified particles. To isolate the $B^{\pm} \to K^{\pm}K^{\mp}K^{\pm}$ signal, some pre-selection cuts must be applied. Specifically, we exploit Particle Identification (PID) variables to ensure that all three final-state tracks are highly likely to be kaons. The applied criteria require a high kaon probability ($P(K) > 0.5$), a low pion probability ($P(\pi) < 0.5$), and a strict muon veto (rejecting any track with hits in muon stations). The distributions of these probability variables, and the effect of the applied cuts, are illustrated in Figure 4.

After applying this PID pre-selection, the invariant mass of the B meson candidates in the real dataset is reconstructed. The resulting distribution is shown in Figure 5, where we can see that, while the B meson signal peak remains distinctly visible and well-defined, a continuous background is now present. Notably, an accumulation of background events is visible in the lower invariant mass region (below 5200 MeV/c^2). This is primarily caused partially reconstructed decays—events where one or more final-state particles (such as a neutral pion or a photon) were not detected, resulting in a lower reconstructed invariant mass.



(a) Kaon probability $P(K)$. Candidates with $P(K) < 0.5$ are rejected.



(b) Pion probability $P(\pi)$. Candidates with $P(\pi) > 0.5$ are rejected.

Figure 4: Particle identification probability distributions for the final-state tracks. The applied kinematic cuts are designed to suppress combinatorial background and misidentified particles.

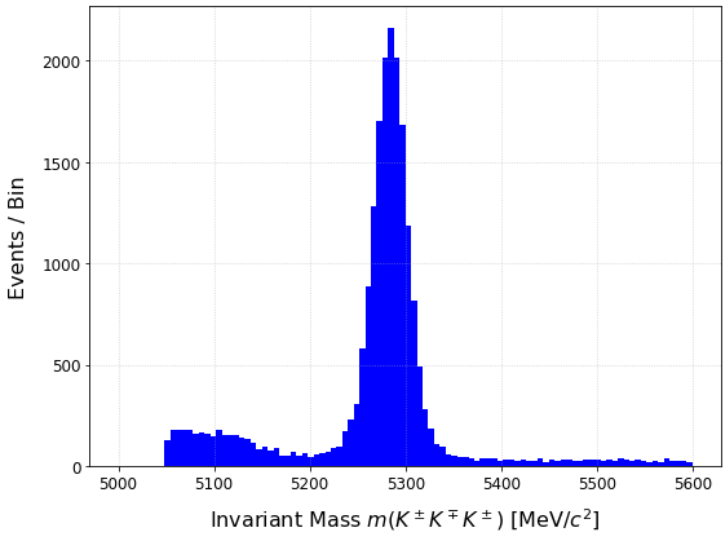


Figure 5: Reconstructed invariant mass distribution for the three kaons system, with the real data from the LHCb detector.

3.3 Global CP Asymmetry Search

Following the invariant mass reconstruction, the analysis proceeded with the search for global CP violation. The primary observable for this study is the global CP asymmetry (A_{CP}) between B^+ and B^- decays, defined as:

$$A_{CP} = \frac{N^+ - N^-}{N^+ + N^-} \tag{2}$$

where N^- and N^+ represent the total number of observed B^- and B^+ decays, respectively. The parent meson charge was determined on an event-by-event basis by summing the charges of the three final-state kaons.

The statistical uncertainty associated with this asymmetry (σ_{stat}) is calculated using the following formula:

$$\sigma_{\text{stat}} = \sqrt{\frac{1 - A_{CP}^2}{N^- + N^+}} \tag{3}$$

In addition to the statistical error, a 1% systematic uncertainty (σ_{syst}) was taken into account for the asymmetric production cross-section of B^+ and B^- mesons in the collisions at the LHC[1]. To obtain the total uncertainty of the measurement (σ_{tot}), the statistical and systematic errors are combined in quadrature:

$$\sigma_{\text{tot}} = \sqrt{\sigma_{\text{stat}}^2 + \sigma_{\text{syst}}^2} \tag{4}$$

Finally, the statistical significance (S) of the measurement is evaluated as the ratio of the absolute asymmetry to its total uncertainty, $S = |A_{CP}|/\sigma_{\text{tot}}$. In particle physics, a significance exceeding 3 standard deviations (3σ) constitutes "evidence" of an effect, whereas 5σ or more is required to claim a formal "observation" or "discovery".

The results of the global CP asymmetry measurement are summarized in Table 1.

N^+	N^-	A_{CP}	σ_{stat}	σ_{syst}	σ_{tot}	Significance (S)
12390	11505	0.037	0.006	0.010	0.012	3.1 σ

Table 1: Summary of the global CP asymmetry measurement. The total uncertainty σ_{tot} combines the statistical error and the 1% systematic error in quadrature.

As reported in Table 1, the measurement yields a global CP asymmetry of $A_{CP} = 0.037 \pm 0.012$. The positive sign indicates a higher decay rate for the matter transition ($B^+ = (u\bar{b})$) compared to the antimatter one ($B^- = (\bar{u}b)$)

[1]This value is an estimate, and comes from the fact that the LHC is a proton-proton collider, not a proton-antiproton one. Also, the same argument can be said for the detector: since it is made of matter, antimatter particle might interact differently with the detector's active parts.

in this specific channel. We calculated a significance of 3.1σ, which exceeds the threshold required in particle physics to claim evidence of a physical phenomenon, but we wnat to go further. The next phase of our study investigates more thoroughly the decay dynamics through Dalitz plots, aiming to identify specific phase-space regions where local CP violation is significantly enhanced.

3.4 Dalitz Plots and Local CP Asymmetry

The decay of a B meson into a three-body final state can proceed either directly into three particles or via an intermediate unstable particle. For instance, the decay $B^+ \to K^+K^+K^-$ may occur through an intermediate neutral resonance R^0, such that $B^+ \to K^+R^0$, followed immediately by the decay $R^0 \to K^+K^-$. Due to energy and momentum conservation, the momenta of the three final-state particles are not independent. Consequently, the kinematics of a three-body decay can be completely described by exactly two independent variables. Conventionally, these variables are chosen to be the squared invariant masses of two pairs of the daughter particles. A Dalitz plot exploits this phase-space property by plotting one squared invariant mass against another. In this two-dimensional representation, decays proceeding through specific intermediate resonant states manifest as densely populated horizontal, vertical, or diagonal bands.

In our search for CP violation, Dalitz plots are a crucial tool. The primary goal of this analysis is to study charmless decays, which are mediated by the rare $b \to u$ quark transitions where CP violation is expected to be prominent. However, the dataset is contaminated by events proceeding via dominant $b \to c$ transitions, which produce intermediate charm resonances (such as the D^0 meson). By mapping the squared invariant masses of the kaon pairs (m^2_{12} and m^2_{13}) onto a Dalitz plot, we can visually identify these charm-mediated background resonances as distinct bands and apply targeted kinematic cuts to exclude them, isolating the pure charmless signal. The results of this procedure before any background resonance veto are shown in Figure 6, displaying the phase-space distributions for different pairs of invariant masses.

To further improve the visibility of the intermediate resonances, the Dalitz plot variables can be redefined. Since the two possible intermediate kaon pairs (K^+K^-) consist of identical particle types, their kinematic distributions are inherently symmetric. By imposing a mass ordering, we sort the two squared invariant masses event-by-event into a higher mass pair (m^2_{High}) and a lower mass pair (m^2_{Low}). This procedure effectively "folds" the Dalitz plot along its symmetry axis. As a result, the total phase space area is halved while retaining the exact same number of events, which doubles the statistical density and reveals the resonant structures much more clearly (Figure 7).

In this folded representation, the charm background resonances appear as distinct horizontal and vertical bands. Specifically, the band corresponding to the D^0 meson is clearly visible around $m^2_{D^0} \approx 3.48 \times 10^6$ MeV$^2/c^4$. Additional traces from charmonium states, such as the J/ψ and χ_{c0}, can also be identified. Since our study focuses exclusively on charmless decays, these specific mass regions were from the dataset. The resulting pure charmless Dalitz plot, divided

removed

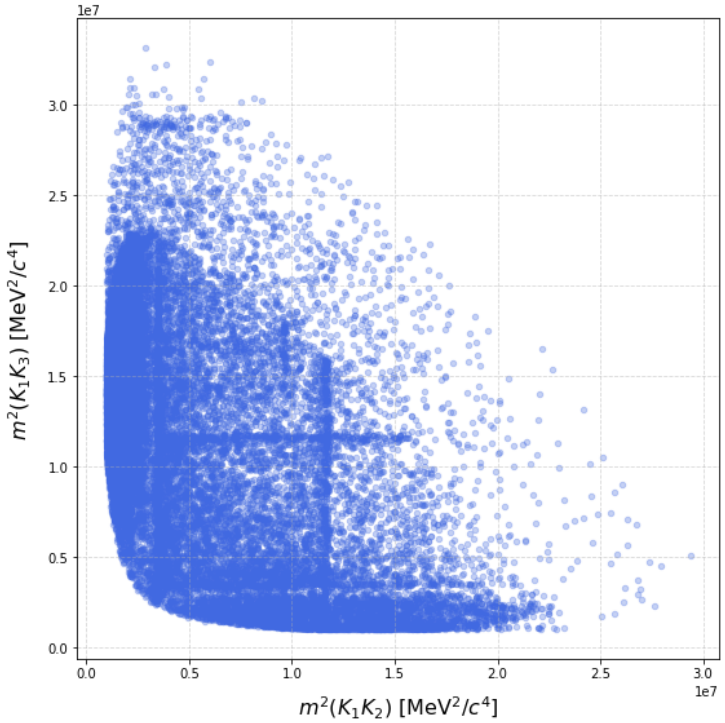


Figure 6: Dalitz plot of m^2_{12} vs m^2_{13} for the $B^{\pm} \to K^{\pm}K^{\mp}K^{\pm}$ candidates before applying resonance cuts. Intermediate resonant states are visible as high-density bands across the phase space.

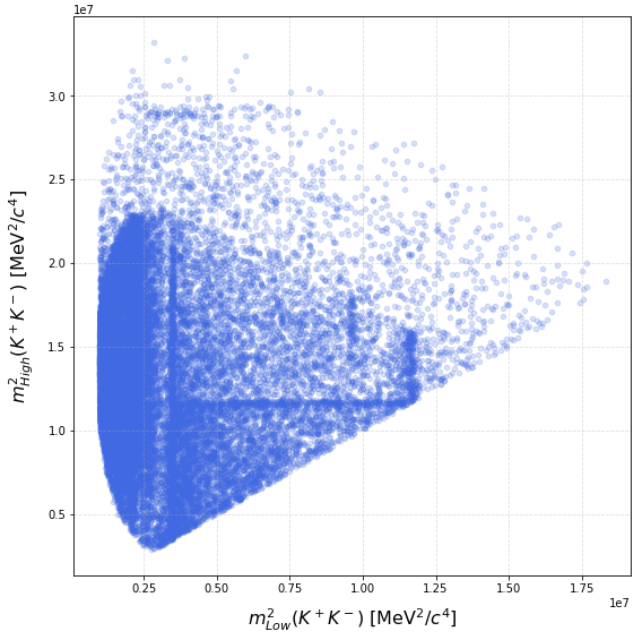


Figure 7: Folded Dalitz plot for the $B^{\pm} \to K^{\pm}K^{\mp}K^{\pm}$ decays. The mass ordering (m^2_{High} vs m^2_{Low}) increases the event density, enhancing the visibility of intermediate resonances.

into distinct resonant regions, is shown in Figure 8.

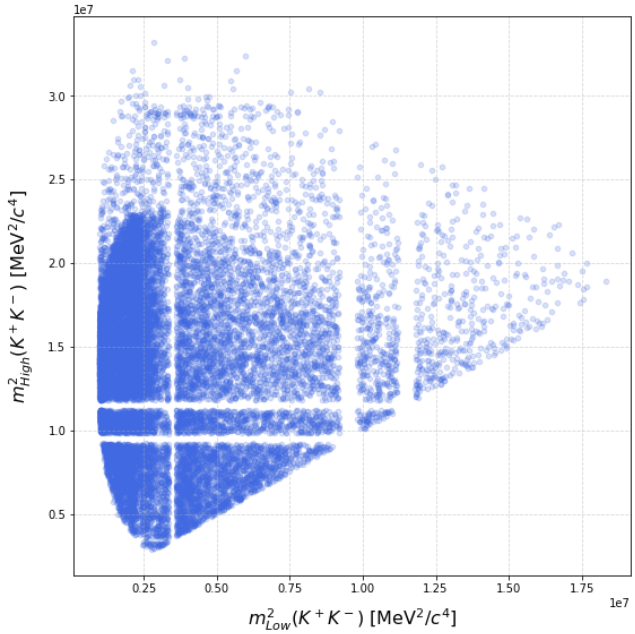


Figure 8: Cleaned folded Dalitz plot after applying kinematic cuts to remove the D^0, J/ψ, and χ_{c0} charm resonances.

With the background successfully removed, the dataset is split into B^+ and B^- decays to search for local CP violation. The Dalitz plot is partitioned into a two-dimensional grid, and the CP asymmetry (A_{CP}) is calculated bin-by-bin (Figure 9). However, observing a large measured asymmetry in a specific bin does not immediately imply genuine CP violation, as bins with very few events are subject to large statistical fluctuations and may still be compatible with zero. To account for this, the statistical uncertainty (σ_A) is computed for each bin using the same formulas applied in the global analysis. By dividing from the absolute asymmetry its uncertainty, we obtain the significance in each bin. Plotting this significance provides a way to select specific phase space regions with CP violation (Figure 10).

Basing on the significance plot, we isolated the specific kinematic region with definitive evidence of local CP violation: $m^2_{\text{Low}} \in [0.0, 3.5 \cdot 10^6]$ and $m^2_{\text{High}} \in [5.0 \cdot 10^6, 25 \cdot 10^6]$ MeV$^2/c^4$. To visually confirm the signal discrepancy, the invariant mass distributions for the B^+ and B^- candidates specifically for the events falling within this high-significance area are shown again, in Figure 11.

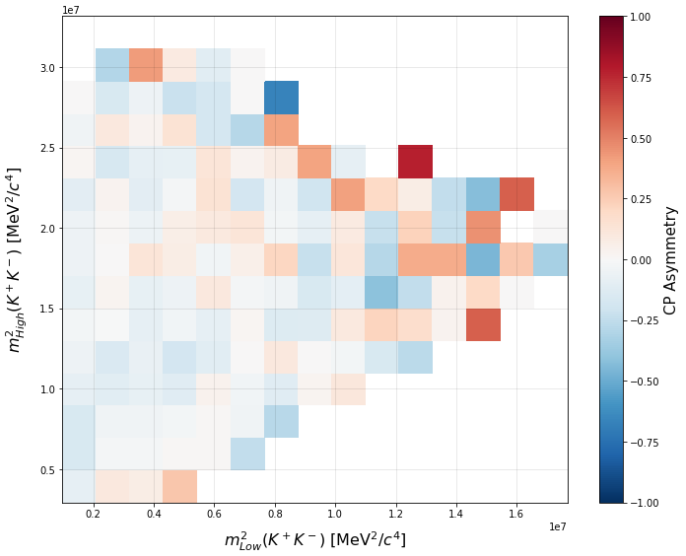


Figure 9: Distribution of the local CP asymmetry (A_{CP}) across the folded Dalitz plot phase space. White or empty areas represent kinematic regions with insufficient statistics (below the minimum event threshold).

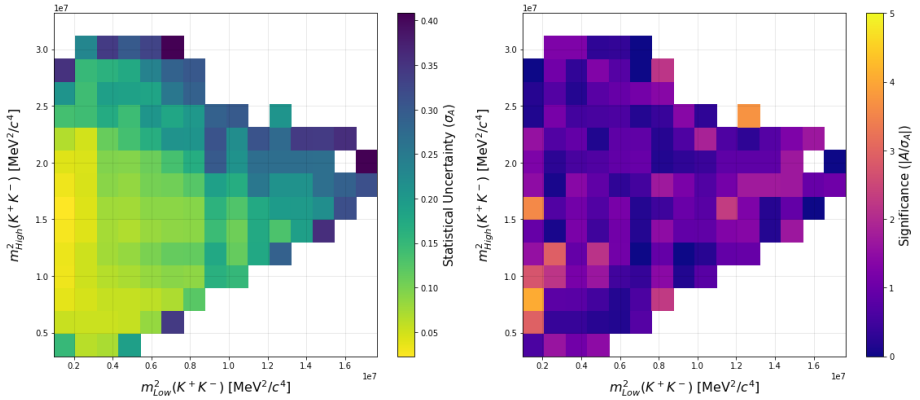


Figure 10: 2D histograms of the statistical uncertainty (left) and the resulting significance (right) of the local CP asymmetry. Bins with a significance greater than 5σ indicate regions of clear CP violation.

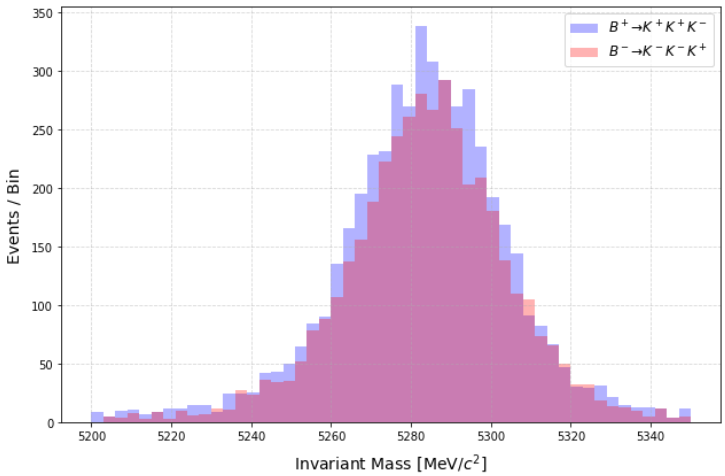


Figure 11: Invariant mass distributions of the three-kaon final state for B^+ (blue) and B^- (red) candidates, with the data from the high-significance kinematic region of the Dalitz plot.

Finally, the CP asymmetry and its statistical significance are re-calculated exclusively for this kinematic region. Also, the *global* CP asymmetry was re-computed using the dataset without the charm resonances. The results are summarized in Table 2 and 3.

N^+	N^-	Local A_{CP}	Uncertainty (σ_A)	Significance (S)
6279	5626	0.0549	0.0092	5.99 σ

Table 2: Summary of the local CP violation measurement within the high-significance kinematic region.

N^+	N^-	Global A_{CP}	Uncertainty (σ_A)	Significance (S)
10240	9467	0.039	0.012	3.2 σ

Table 3: Summary of the global CP violation measurement using the charmless dataset.

The resulting local asymmetry has a significance of 5.99σ, well above the 5σ threshold required to announce a definitive discovery. This confirms that while global CP violation in this decay channel is relatively small, even after the computation with the charmless dataset, the local CP violation localized around specific charmless resonances is extremely prominent.

4 Conclusions

In this study, an analysis of $B^{\pm} \to K^{\pm}K^{\mp}K^{\pm}$ decays was performed using data from the LHCb experiment to investigate charge-parity violation. Through rigorous invariant mass reconstruction and targeted particle identification preselection cuts, the charmless B meson signal was successfully isolated from the combinatorial background. Our initial global measurement yielded a CP asymmetry of $A_{CP} = 0.037 \pm 0.012$. With a statistical significance of 3.1σ, this provided strong evidence of an underlying matter-antimatter asymmetry. This confirms that the observed discrepancy is a genuine physical effect rather than a statistical fluctuation. However, this result falls short of the rigorous 5σ standard required to claim a definitive "observation" or discovery of global CP violation. This limitation motivated the next phase of our study: a further investigation of the decay dynamics through Dalitz plots. So, we performed a targeted phase-space analysis using folded Dalitz plots, which allowed us to identify and remove charm-mediated background resonances (such as the D^0, J/ψ, and χ_{c0}). We then re-calculated the global CP asymmetry, which yielded a slightly bigger value of $A_{CP} = 0.039 \pm 0.012$, with a statistical significance of 3.2σ. Finally, we associated the CP asymmetry across different kinematic regions of the decay. This approach proved to be highly effective because isolating a specific high-significance kinematic region ($m^2_{\text{Low}} \in [0.0, 3.5 \times 10^6]$ and $m^2_{\text{High}} \in [5.0 \times 10^6, 25 \times 10^6]$ MeV$^2/c^4$), we measured a local CP asymmetry of $A_{CP} = 0.0549 \pm 0.0092$. This localized measurement dramatically increased the statistical significance to 5.99σ, comfortably surpassing the threshold required to claim a formal observation.

Ultimately, this study confirms the presence of CP violation in $B^{\pm} \to K^{\pm}K^{\mp}K^{\pm}$ decays and also demonstrates the power of Dalitz plot analysis. Although CP violation in the quark sector alone cannot account for the observed matter–antimatter asymmetry of the Universe, the results of this study are consistent with CP violation effects predicted by the Standard Model in allowed decay processes.

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